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Yuh

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(54) **FERROELECTRIC FIELD-EFFECT TRANSISTOR (FeFET) MEMORY**

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(60) Provisional application No. 63/041,515, filed on Jun. 19, 2020.

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GI1C 11/22 (2006.01)
GI1C 11/56 (2006.01)
H10B 51/00 (2023.01)

(52) **U.S. Cl.**
CPC **GI1C 11/223** (2013.01); **GI1C 11/2259** (2013.01); **GI1C 11/2273** (2013.01); **GI1C 11/2275** (2013.01); **GI1C 11/5657** (2013.01); **H10B 51/00** (2023.02)

(58) **Field of Classification Search**
CPC **GI1C 11/223**; **GI1C 11/2259**; **GI1C 11/2275**; **GI1C 11/5657**
See application file for complete search history.

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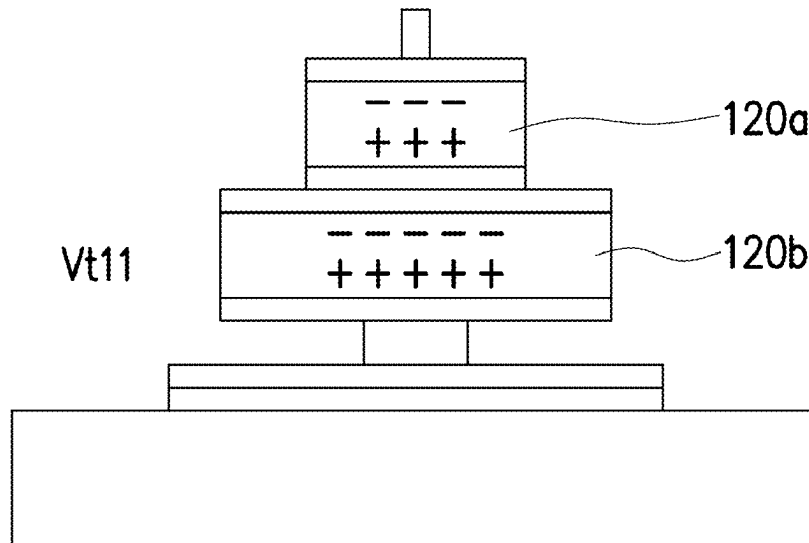
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(57) **ABSTRACT**

A memory device includes a plurality of memory cells. Each memory cell includes a multi-gate FeFET that has a first source/drain terminal, a second source/drain terminal, and a gate with a plurality of ferroelectric layers configured such that each of the ferroelectric layers has a respective unique switching E-field.

20 Claims, 13 Drawing Sheets



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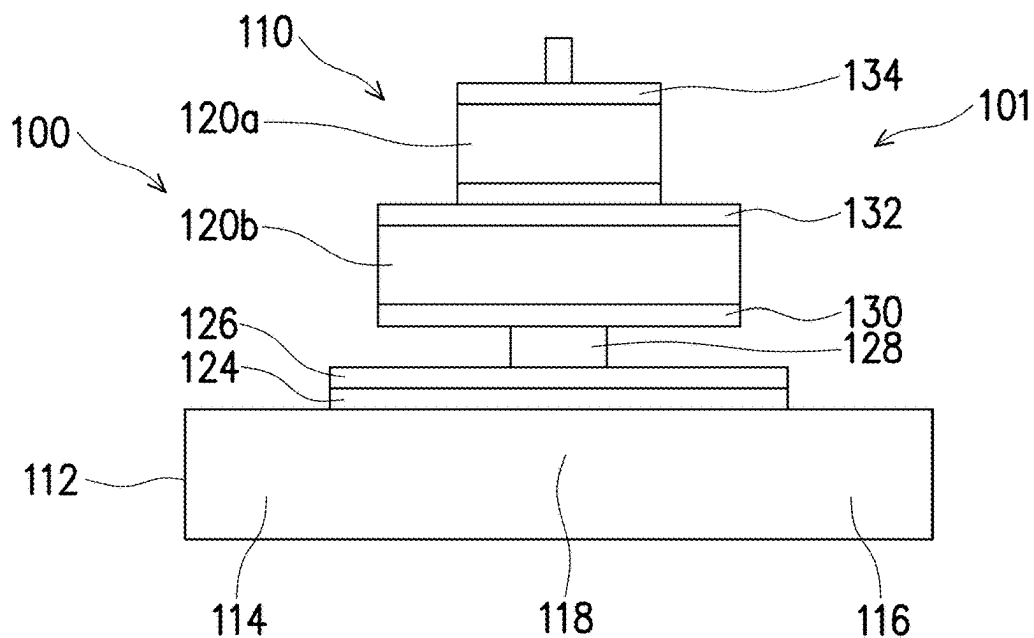


FIG. 1A

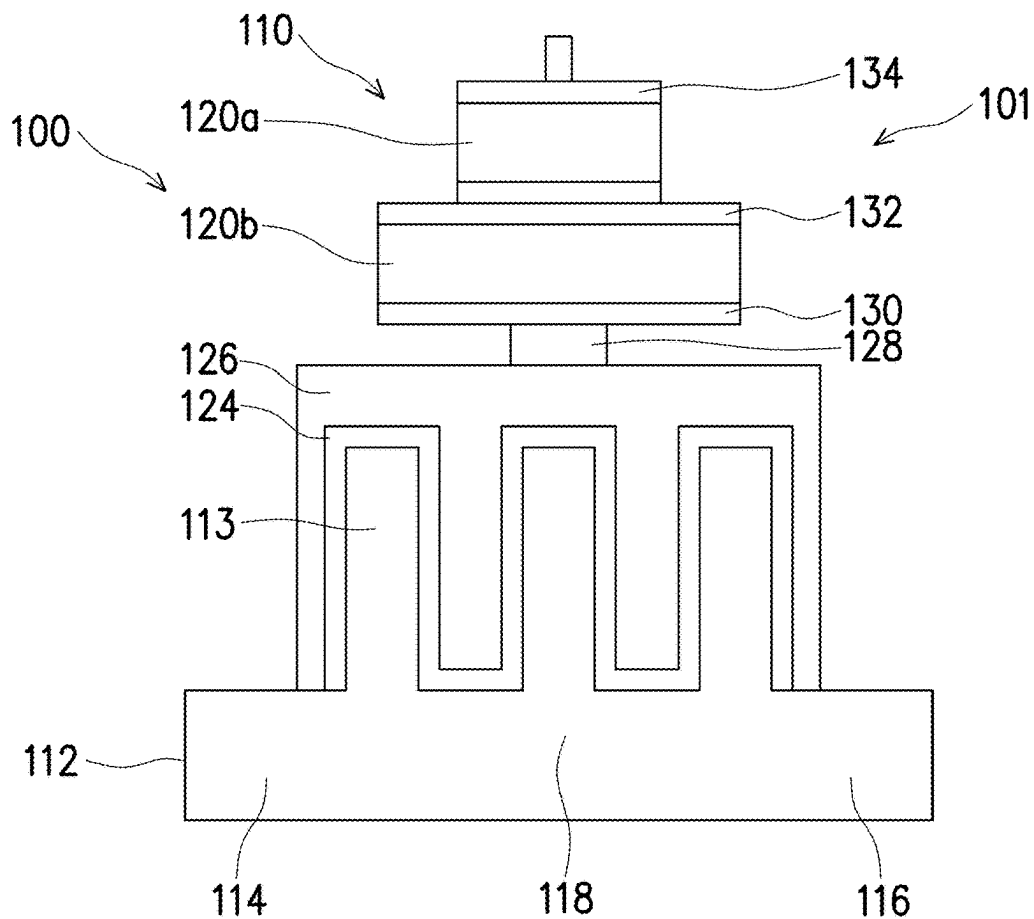


FIG. 1B

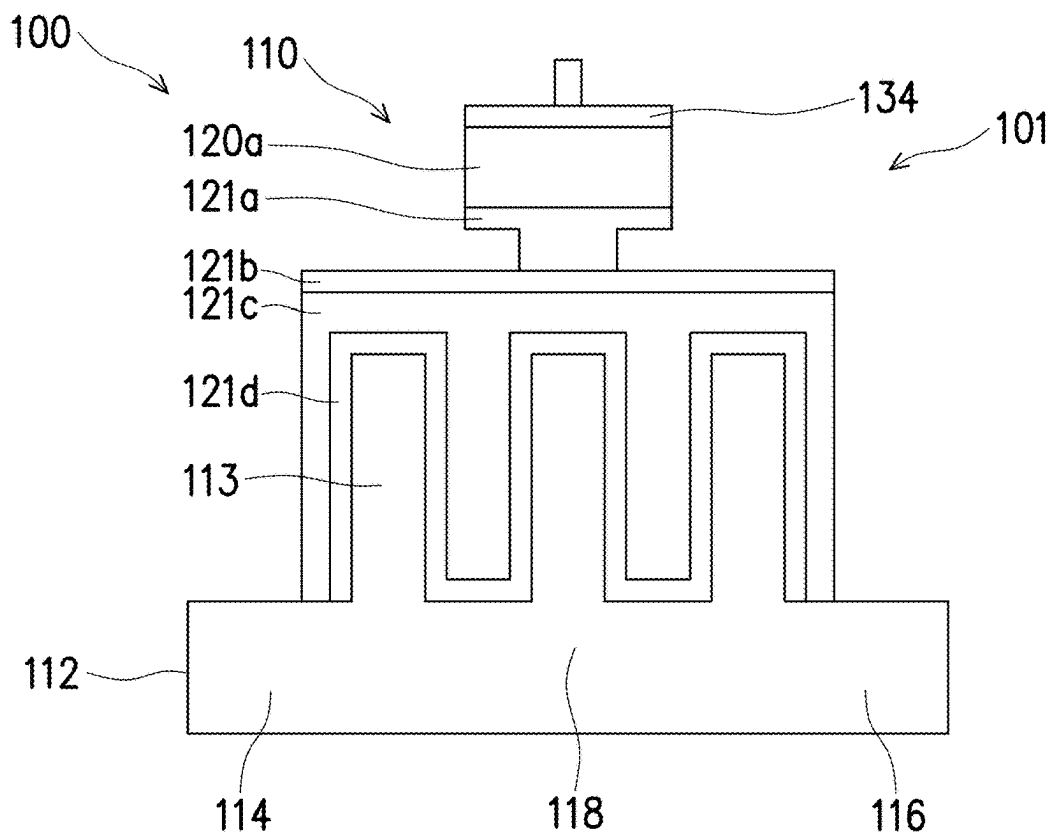


FIG. 1C

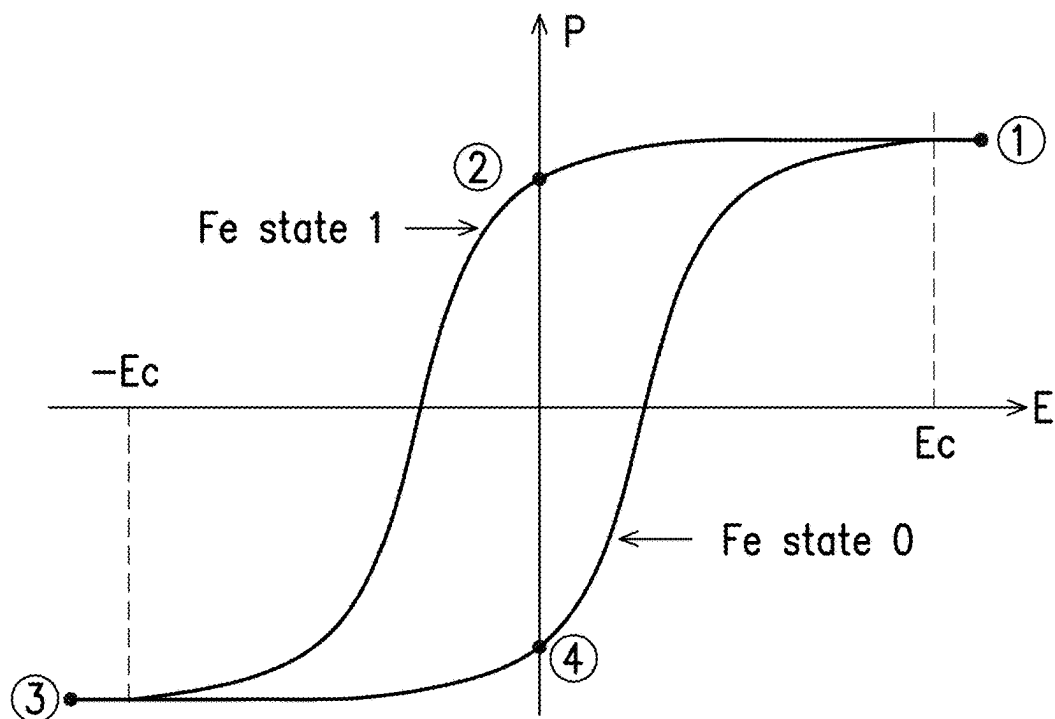


FIG. 2

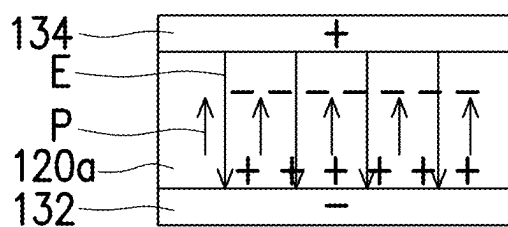


FIG. 3

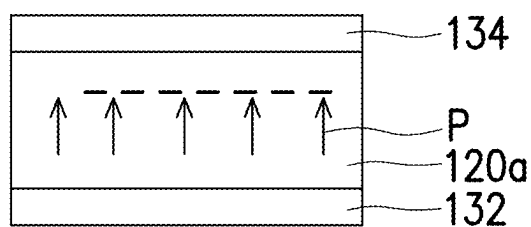


FIG. 4

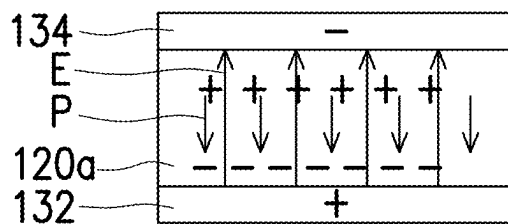


FIG. 5

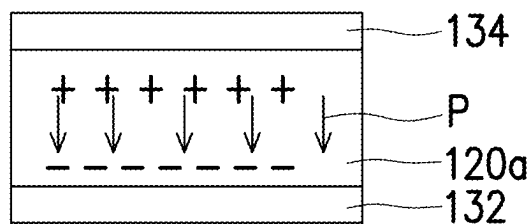


FIG. 6

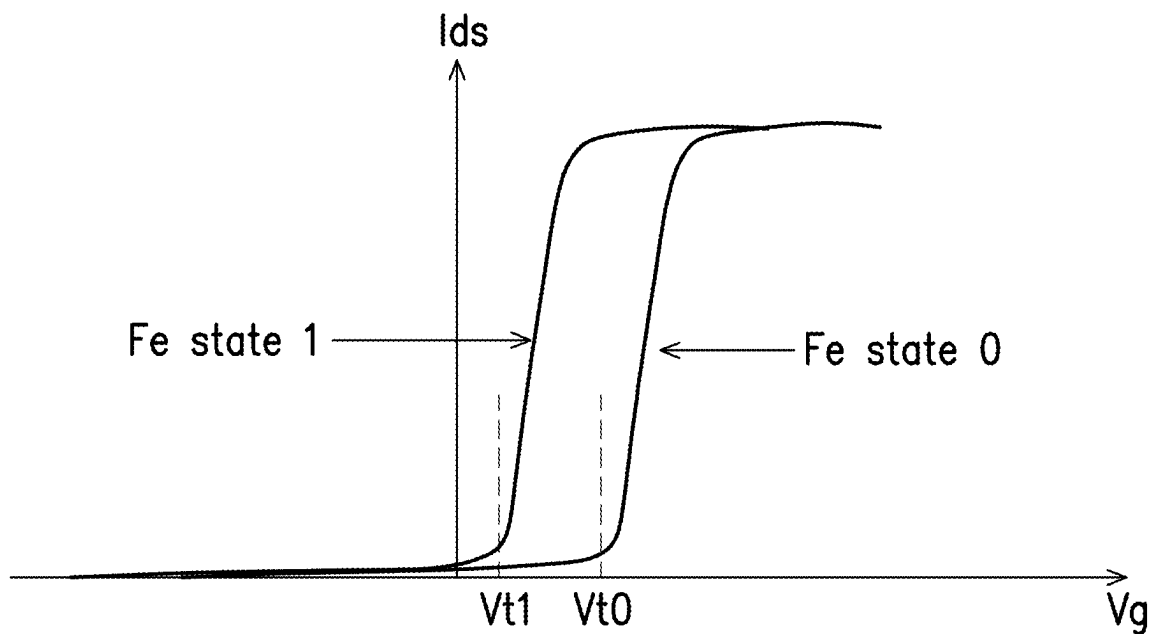


FIG. 7

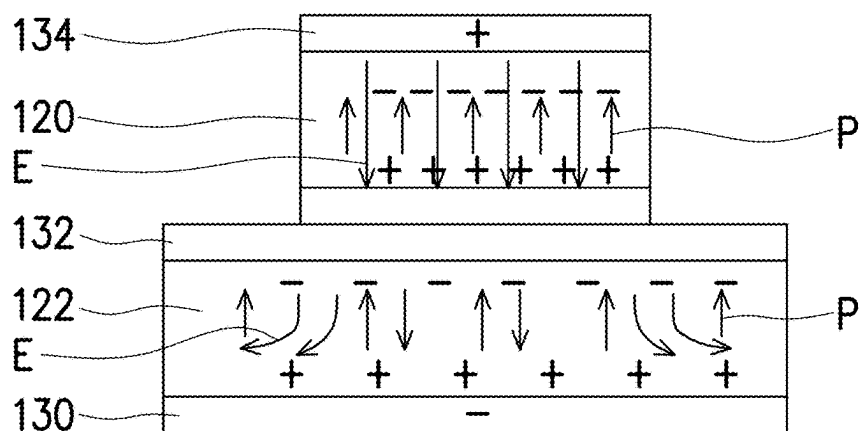


FIG. 8

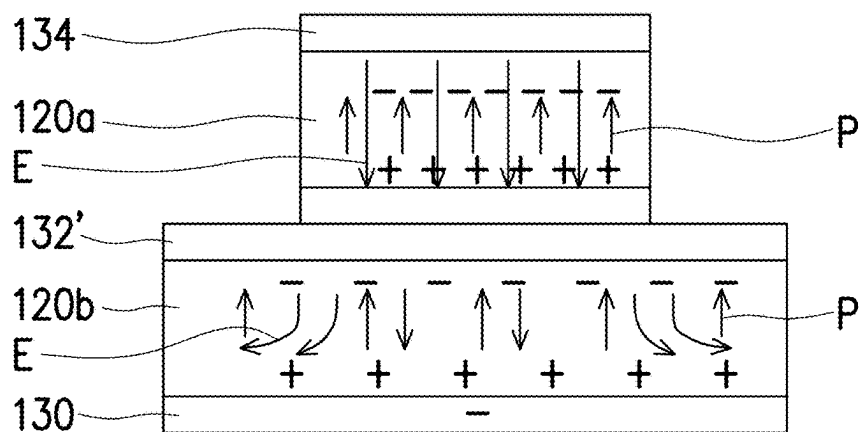


FIG. 9

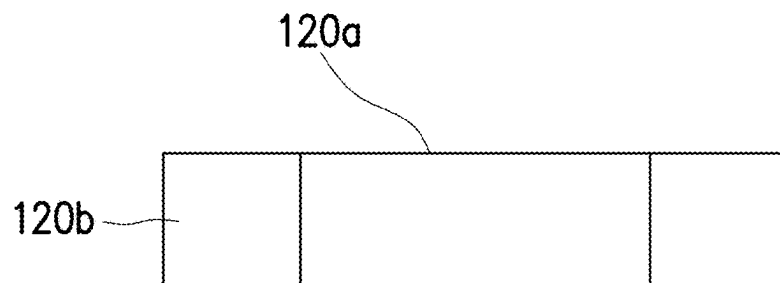


FIG. 10

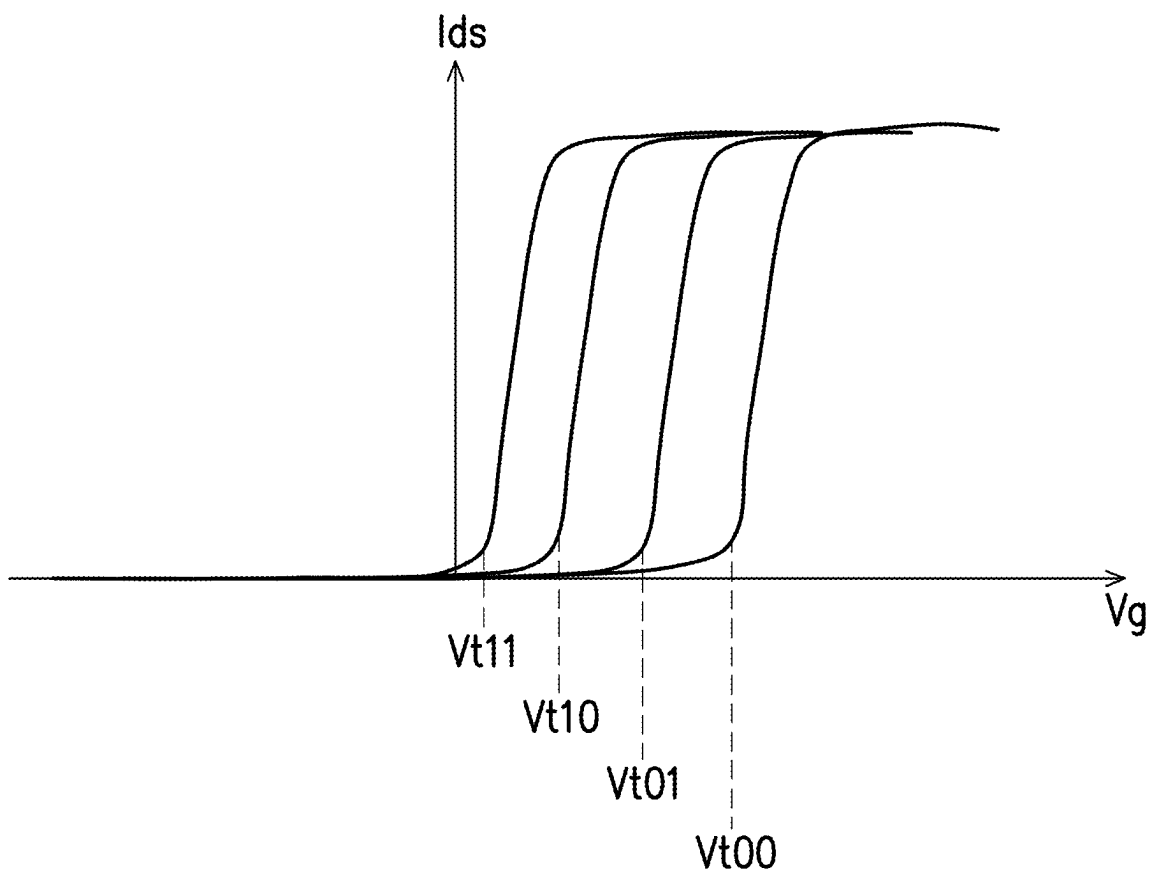


FIG. 11

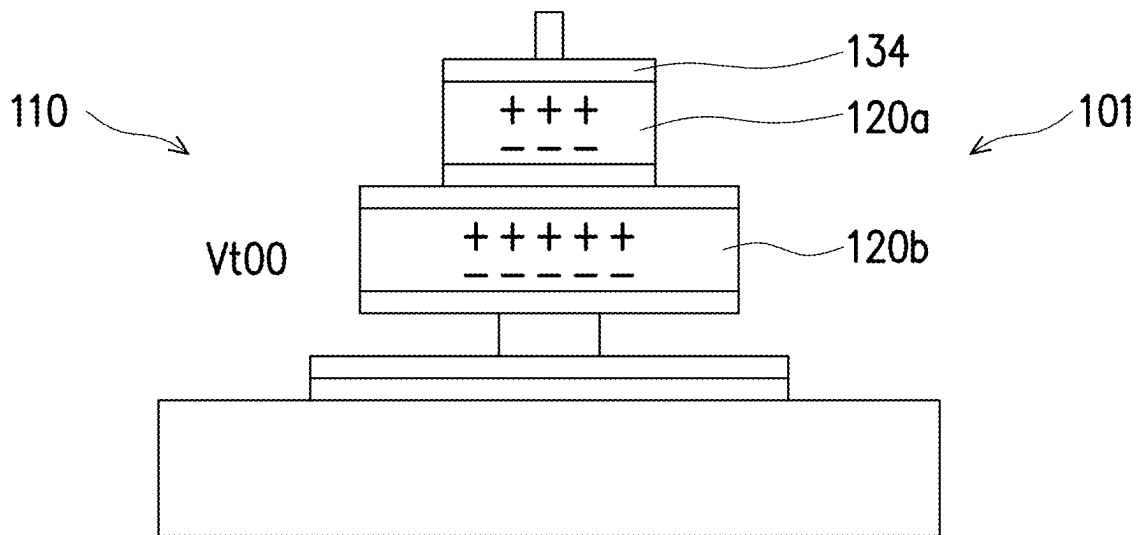


FIG. 12

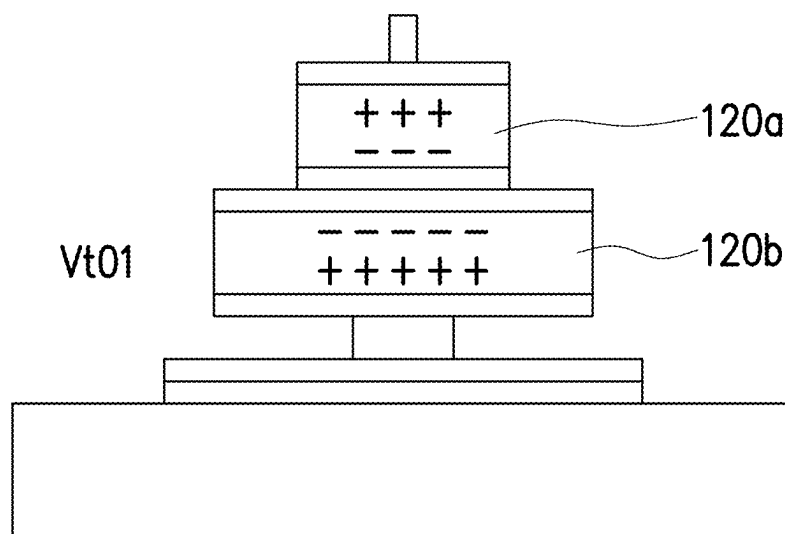


FIG. 13

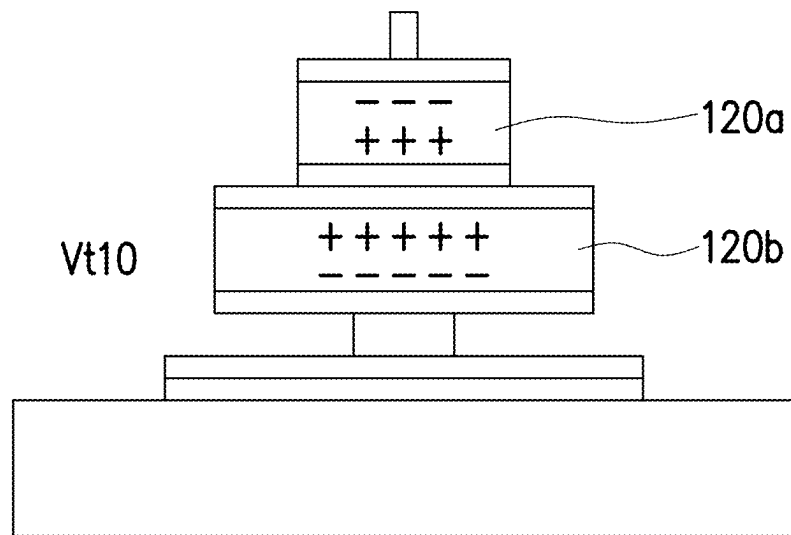


FIG. 14

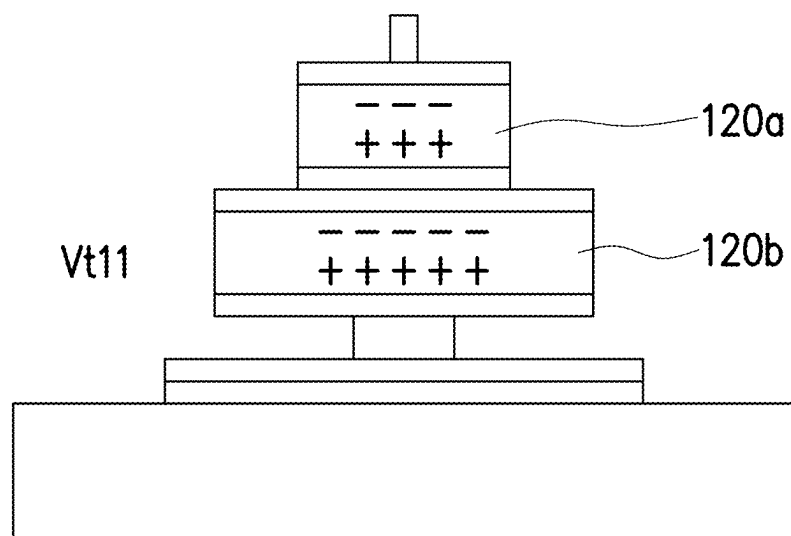


FIG. 15

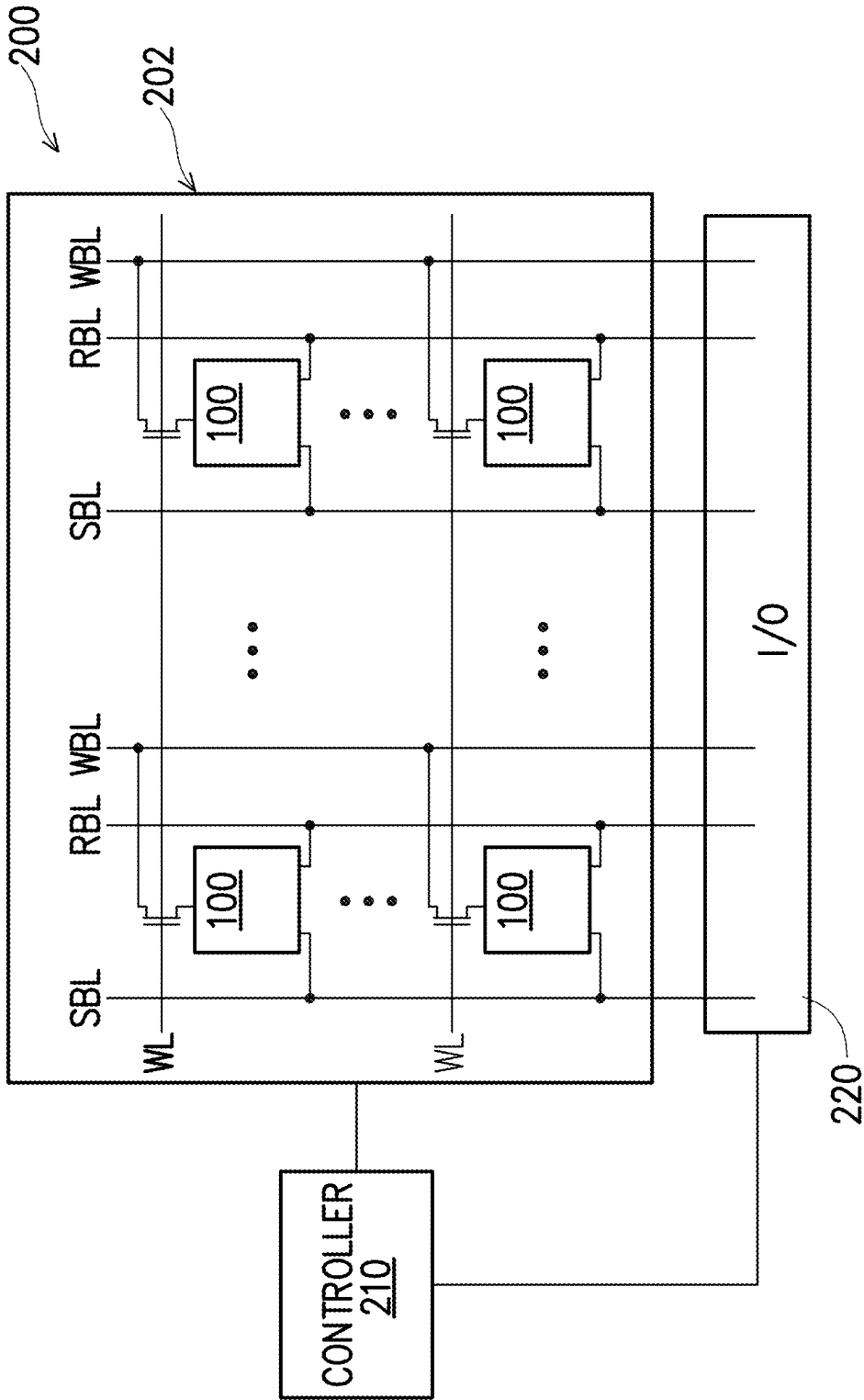


FIG. 16

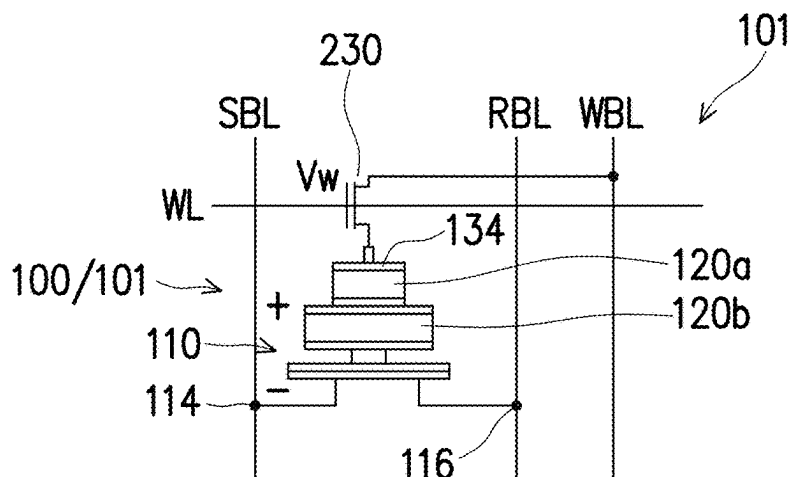


FIG. 17

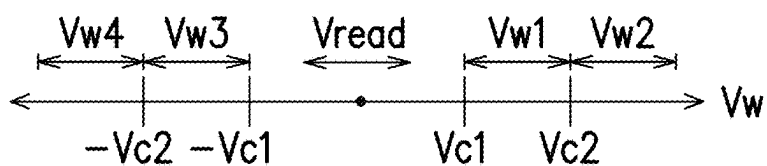


FIG. 18

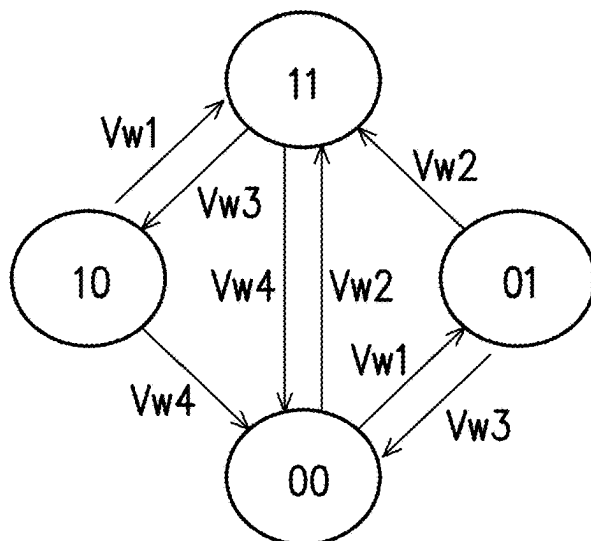


FIG. 19

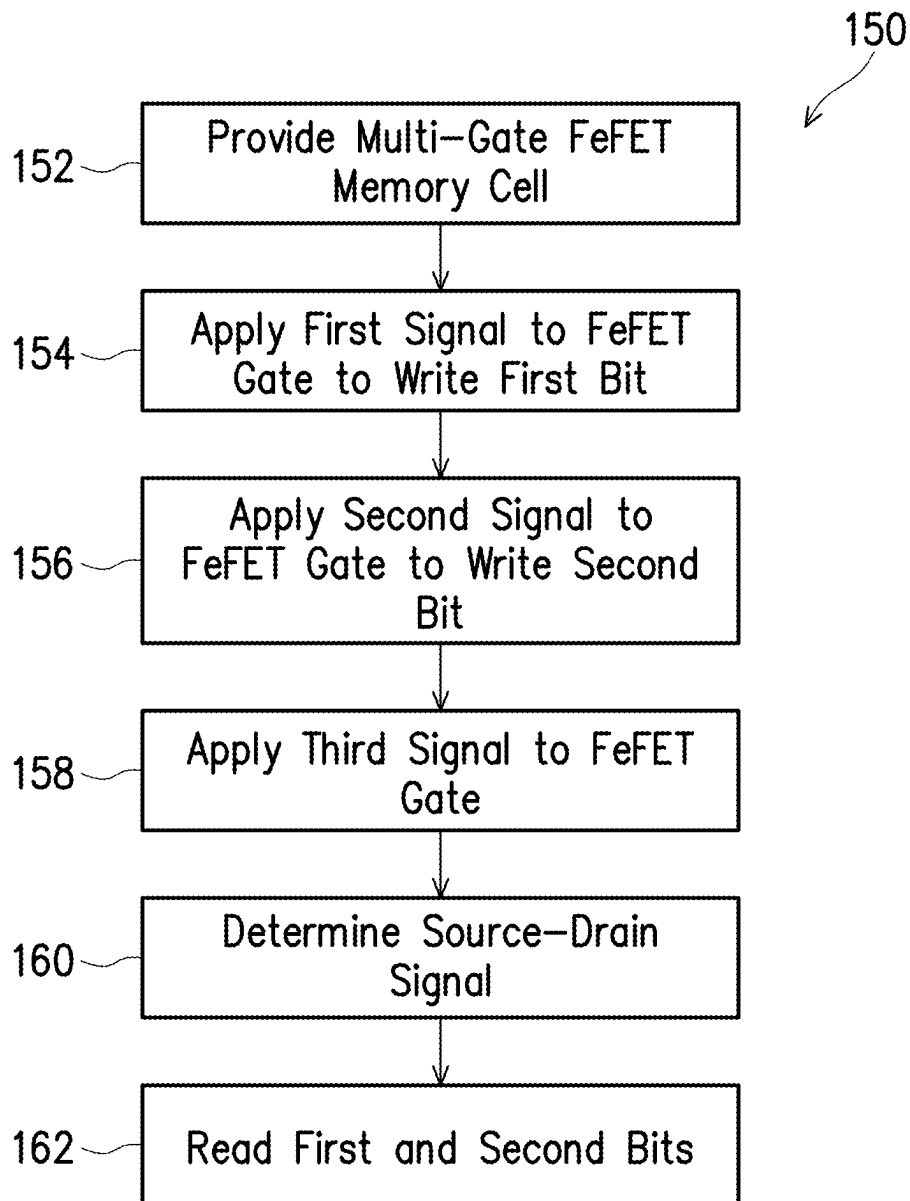


FIG. 20

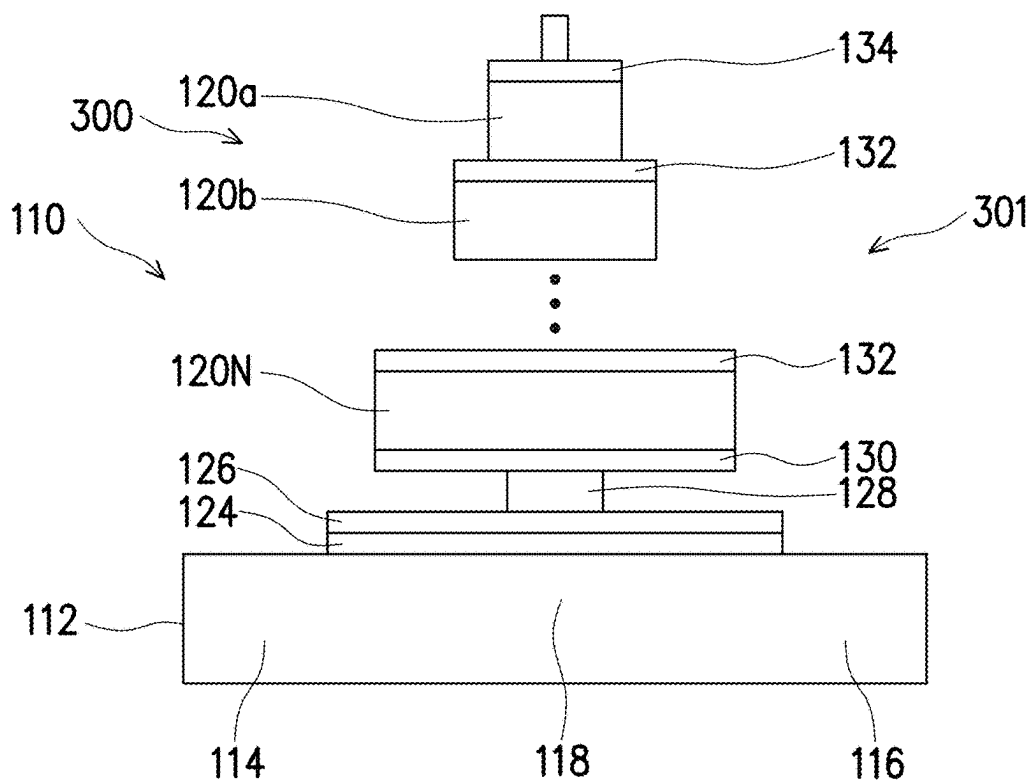


FIG. 21

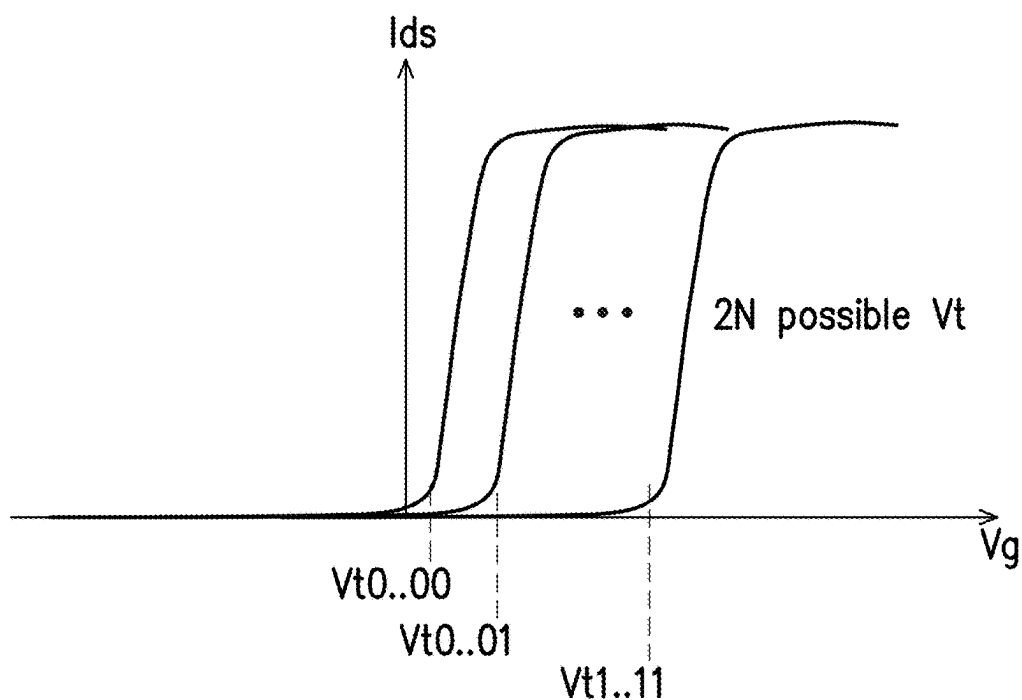


FIG. 22

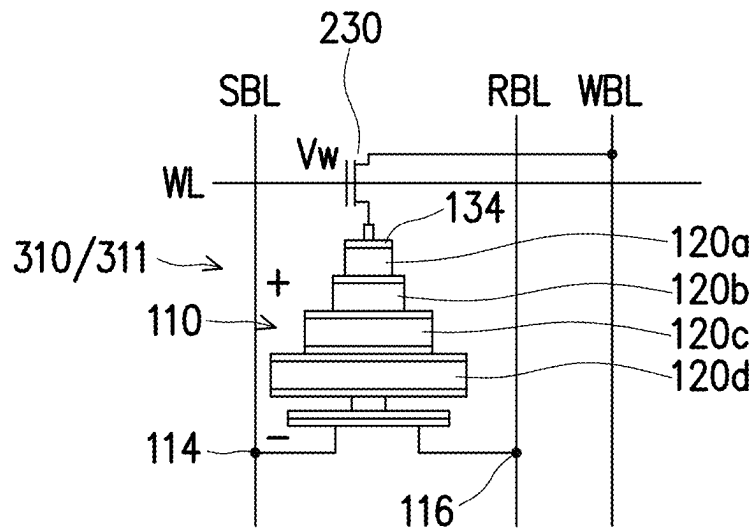


FIG. 23

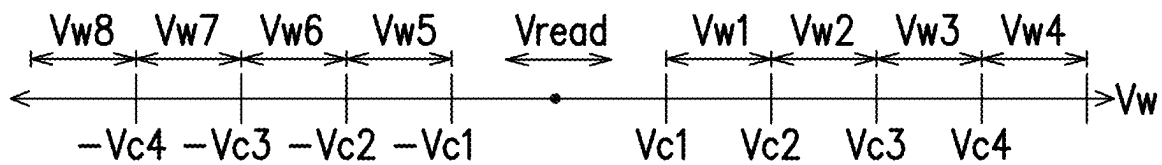


FIG. 24

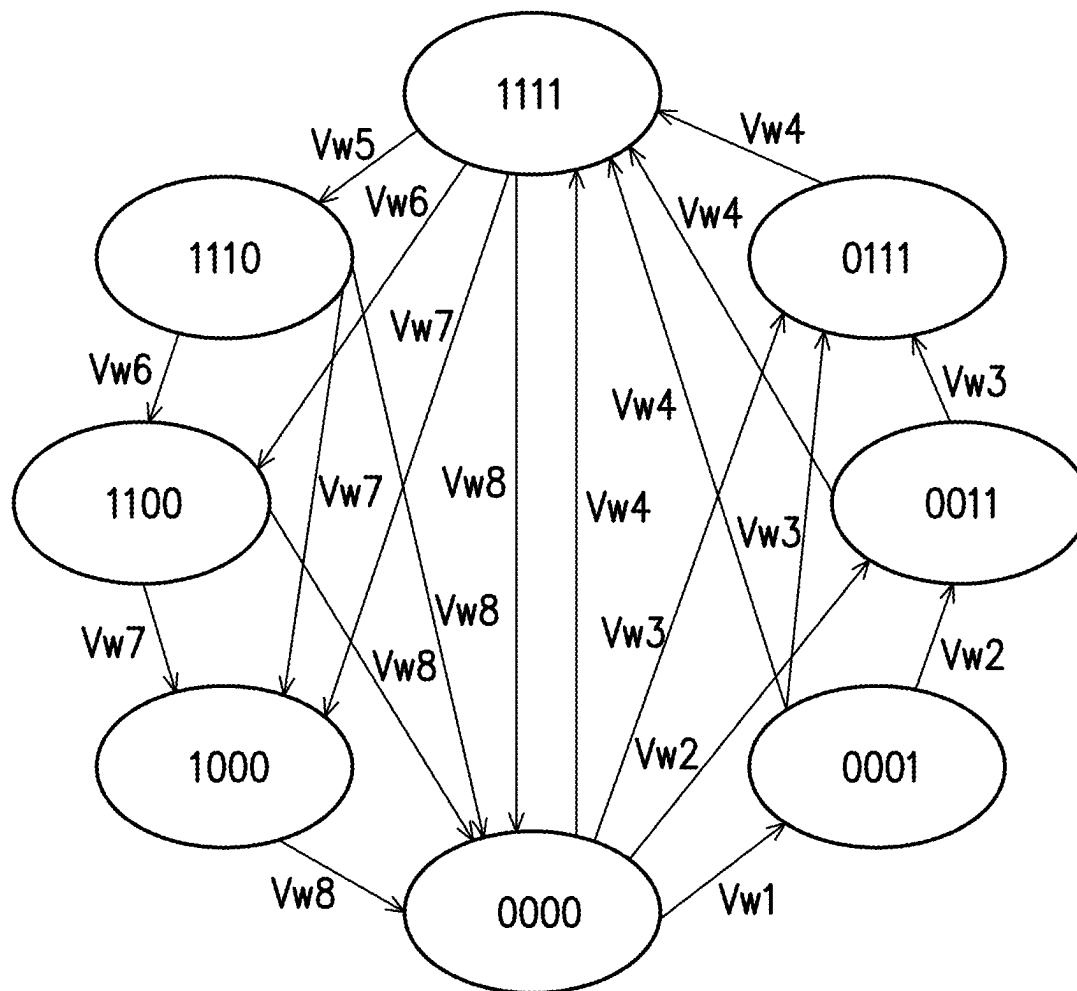


FIG. 25

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FERROELECTRIC FIELD-EFFECT TRANSISTOR (FeFET) MEMORY

CROSS REFERENCE TO RELATED APPLICATION

This application is a continuation of Ser. No. 17/816,143, filed on Jul. 29, 2022, which is a continuation of U.S. patent application Ser. No. 17/229,194, now U.S. Pat. No. 11,450,370, filed Apr. 13, 2021, which claims the benefit of U.S. Provisional Application No. 63/041,515, filed Jun. 19, 2020, and titled “Ferroelectric Field-Effect Transistor (FeFET) Memory,” the disclosures of which are hereby incorporated herein by reference.

BACKGROUND

A ferroelectric field-effect transistor (FeFET) is a type of field-effect transistor that includes a ferroelectric material sandwiched between the gate electrode and source-drain conduction region of the device. Permanent electrical field polarization in the ferroelectric causes this type of device to retain the transistor's state (on or off) in the absence of an electrical bias. FeFET based devices are used in FeFET memory or FeRAM.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion. In addition, the drawings are illustrative as examples of embodiments of the invention and are not intended to be limiting.

FIG. 1A is a block diagram schematically illustrating an example of a multi-bit memory cell in accordance with disclosed embodiments.

FIG. 1B is a block diagram schematically illustrating alternative aspects of the multi-bit memory cell shown in FIG. 1A in accordance with disclosed embodiments.

FIG. 1C is a block diagram schematically illustrating alternative aspects of the multi-bit memory cells shown in FIGS. 1A and 1B in accordance with disclosed embodiments.

FIG. 2 is a chart illustrating a hysteresis curve for ferroelectric material for polarization induced by an applied electrical field in accordance with disclosed embodiments.

FIGS. 3-6 are block diagrams of a ferroelectric gate structure illustrating polarization states shown in FIG. 2 in accordance with disclosed embodiments.

FIG. 7 is a chart illustrating two threshold voltage (V_t) levels for a single ferroelectric gate layer of a multiple gate FeFET in accordance with disclosed embodiments.

FIG. 8 is a block illustrating a gate structure including two ferroelectric gate layers of a multiple gate FeFET in accordance with disclosed embodiments.

FIG. 9 is a block illustrating an alternative gate structure including two ferroelectric gate layers of a multiple gate FeFET in accordance with disclosed embodiments.

FIG. 10 is a top view illustrating aspects of the gate structure shown in FIGS. 8 and 9 in accordance with disclosed embodiments.

FIG. 11 is a chart illustrating V_t levels for a dual gate FeFET in accordance with disclosed embodiments.

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FIGS. 12-15 are block diagrams illustrating portions of the multi-bit memory cell shown in FIG. 1A, showing the polarizations corresponding to the V_t levels shown in FIG. 11 in accordance with disclosed embodiments.

FIG. 16 is a schematic diagram illustrating an example of a memory device that includes multi-bit memory cells such as those shown in FIGS. 1A-1C in accordance with disclosed embodiments.

FIG. 17 is a block diagram illustrating further aspects of the memory device shown in FIG. 16 in accordance with disclosed embodiments.

FIG. 18 is a chart illustrating example relationships among memory cell write voltages and minimum switching voltages in accordance with disclosed embodiments.

FIG. 19 is a chart illustrating example switching transitions of ferroelectric layers corresponding to write voltages shown in FIG. 18 in accordance with disclosed embodiments.

FIG. 20 is a flow diagram illustrating an example method for operating a multi-bit memory cell in accordance with disclosed embodiments.

FIG. 21 is a block diagram schematically illustrating a generalized example of a multi-bit memory cell in accordance with disclosed embodiments.

FIG. 22 is a chart illustrating V_t levels for the generalized multi-bit memory cell of FIG. 21 in accordance with disclosed embodiments.

FIG. 23 is a block diagram schematically illustrating a quadruple gate multi-bit memory cell in accordance with disclosed embodiments.

FIG. 24 is a chart illustrating example relationships among memory cell write voltages and minimum switching voltages for the multi-bit memory cell of FIG. 23 in accordance with disclosed embodiments.

FIG. 25 is a chart illustrating example switching transitions of ferroelectric layers corresponding to write voltages shown in FIG. 24 in accordance with disclosed embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90

degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

A ferroelectric field-effect transistor (FeFET) is a type of field-effect transistor that includes a ferroelectric material sandwiched between the gate electrode and source-drain conduction region of the device. Permanent electrical field polarization in the ferroelectric causes this type of device to retain the transistor's state (on or off) in the absence of an electrical bias. FeFET based devices are used in FeFET memory or FeRAM, among other things.

The ferroelectric material generally replaces the gate oxide of the FET. The switching is caused by applying an electrical field via a voltage between the transistor gate and transistor channel. Specially, for n-channel transistors, ferroelectric switching after application of a sufficiently high positive voltage pulse causes a shift of the threshold voltage (V_t) to lower threshold voltage values. For p-channel transistors a negative voltage pulse causes a shift of the threshold voltage to higher threshold voltage values.

A common type of integrated circuit memory is a static random access memory (SRAM) device. A typical SRAM memory device has an array of memory cells. Each memory cell uses six transistors, for example, connected between an upper reference potential and a lower reference potential (typically ground) such that one of two storage nodes can be occupied by the information to be stored, with the complementary information stored at the other storage node. While SRAM maintains data in the memory array without the need to be refreshed when powered, it is volatile in that data is eventually lost when the memory is not powered. Further, the typical SRAM construction consumes a relatively large area.

FeRAM has generally has a smaller area that may be about 25% smaller in size as compared to conventional SRAM. Some disclosed embodiments provide a memory cell that is capable of storing multiple bits of data, thus further reducing area required for memory cells. More particularly, disclosed examples include a memory cell that has a multi-gate FeFET with a source, a drain and a gate, where the gate has a plurality of ferroelectric layers configured such that each of the ferroelectric layers has a respective unique switching E-field. Some examples may include vertical stacking of ferroelectric layers having different sizes, and thus each of the ferroelectric layers has a different switching E field. This in turn provides a FeFET device that has more V_t values than a conventional FeFET memory cell. Accordingly, multiple level write and multiple level read operations are provided, allowing a memory cell with multiple memory bits that occupies similar area as a single bit FET cell. This allows a reduction of the effective device area per memory bit. Such multiple ferroelectric layers can be provided the back end of the line (BEOL) or middle end of the line (MEOL) processes providing manufacturing flexibility.

FIG. 1A illustrates aspects of a multi-bit memory cell 100 in accordance with disclosed examples. The multi-bit memory cell 100 includes a multi-gate FeFET 101 having a substrate 112 that may be, for example, a semiconductor layer of a semiconductor-on-insulator (SOI) substrate or a bulk semiconductor substrate. The FeFET 101 comprises a pair of source/drain regions 114, 116 that have a first doping type and are arranged within the substrate 112, respectively on opposite sides of a channel region 118. The channel region 118 has a second doping type opposite the first doping and is arranged in the substrate 112 laterally between the

source/drain regions 114, 116. The first and second doping types may, for example, respectively be n-type and p-type, or vice versa.

FIG. 1B illustrates an alternative embodiment in which the substrate 112 defines fins 113 extending therefrom. The source/drain regions 114, 116 have the first doping type and are arranged within the substrate 112, respectively on opposite sides of the channel region 118 having the second doping type opposite the first doping type.

The FeFET 101 includes a gate 110 that has a gate metal layer 126 arranged over the substrate 112 laterally between the source/drain regions 114, 116, and is spaced from the substrate 112 by a non-ferroelectric gate oxide 124. In some examples, the non-ferroelectric gate oxide material is silicon dioxide. The gate electrode 126 is conductive and may comprise metal, doped polysilicon, or a combination thereof. In the illustrated embodiment, the gate electrode 126 is electrically coupled to a BEOL ferroelectric gate structure by a conductive vertical interconnect access (via) 128.

The conductive via 128 may be part of a BEOL interconnect structure arranged over the substrate 112. The interconnect structure may include a multi-layer interconnect (MLI) structure having conductive lines, conductive vias, and/or interposing dielectric layers (e.g., interlayer dielectric (ILD) layers). The interconnect structure may provide various physical and electrical connections, including the gate via 128. The conductive lines may comprise copper, aluminum, tungsten, tantalum, titanium, nickel, cobalt, metal silicide, metal nitride, poly silicon, combinations thereof, and/or other materials possibly including one or more layers or linings. The interposing dielectric layers (e.g., ILD layers) may comprise silicon dioxide, fluorinated silicon glass (FGS), SILK (a product of Dow Chemical of Michigan), BLACK DIAMOND (a product of Applied Materials of Santa Clara, Calif.), and/or other suitable insulating materials. The MLI structure may be formed by suitable processes typical in CMOS fabrication such as CVD, PVD, ALD, plating, spin-on coating, and/or other processes.

The gate 110 of the illustrated FeFET 101 includes a plurality of ferroelectric layers configured such that each of the ferroelectric layers has a respective unique switching E-field. In the illustrate example, the gate 110 includes a first ferroelectric layer 120a and a second ferroelectric layer 120b in a stacked arrangement. A gate metal layer 130 abuts one side of the ferroelectric layer 120b and is connected to the gate electrode 126 by the via 128. A buffer layer 132 is situated between the ferroelectric layer 120a and the ferroelectric layer 120b, with a conductive gate terminal 134 situated on the ferroelectric layer 120a. In alternative embodiments the, ferroelectric layers 120a, 120b are formed directly over the non-ferro gate oxide 124 to form a composite gate structure.

The ferroelectric layers 120a, 120b may be formed in FEOL process layers close to the non-ferroelectric gate oxide 124, or as shown in the example of FIG. 1, the ferroelectric layers 120a, 120b are formed in MEOL or BEOL processes and are connected to the gate metal layer 126 by the via 128. In still further examples, ferroelectric layers 120a, 120b are formed in MEOL or BEOL processes for FinFET, nano sheet, or other gate all around technologies.

In FIGS. 1a and 1b, the ferroelectric layers 120a and 120b are both planar structures. Other embodiments may employ ferroelectric layers having other shapes and arrangements. FIG. 1C illustrates an example of the FeFET 101 that includes the substrate 112 with fin structures 113 as shown

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in the example of FIG. 1B. FIG. 1C further illustrates the planar ferroelectric layer 120, as well as several additional example ferroelectric layers. In addition to the planar ferroelectric layer 120a, FIG. 1C illustrates example ferroelectric layers including a T-shaped ferroelectric layer 121a, a thin planar ferroelectric layer 121b, a thick fin ferroelectric layer 121c, and a thinner ferroelectric layer 121d. For simplicity, other structures of the multi-bit memory cell 100 are omitted in FIG. 1C.

A ferroelectric material is an insulator in which the polarization P induced by an applied electrical field E shows a hysteresis curve as shown in FIG. 2. FIG. 2 shows four states when varying an electrical field applied to a ferroelectric layer. FIGS. 3-6 illustrate the four states shown in FIG. 2 for an example of one of the ferroelectric layers 120a positioned between metal layers such as gate terminals 130, 134 and the buffer layer 132. For simplicity, FIGS. 3-6 only show the first ferroelectric layer 120a between the conductive gate terminal 134 and buffer layer 132. FIG. 3 illustrates a positive E field generated by applying a voltage between the top terminal 134 and the lower terminal 132 as indicated by the downwardly pointing arrows, resulting in the polarization P. State 1 shown in FIG. 2 is a case for writing a data 1, where the illustrated polarization P results from an E field higher than a positive critical electrical field E_c . When the applied E field is removed as shown in FIG. 4, the polarization state remains as shown at state 2 in FIG. 2, i.e. the data high or 1 state. To write a data 0, a negative E field lower than a negative critical field $-E_c$ is applied as shown in FIG. 5, reversing the induced polarity P resulting in state 3 shown in FIG. 2. When the external E field is removed as shown in FIG. 6, the ferroelectric state remains at the 0 state.

Thus, the first ferroelectric layer 120a forming one layer of the gate 110 of the FeFET 101 has two threshold voltage (V_t) levels depending on the polarization of the ferroelectric material. FIG. 7 illustrates two V_t levels for a single ferroelectric gate layer, such as the ferroelectric layer 120a or 120b. As shown in FIG. 7, two distinct V_t values are established that correspond to the two states of the ferroelectric gate material described in conjunction with FIGS. 2-5. The V_{t1} level corresponds to the 1 state shown in FIG. 3, while the V_{t0} level corresponds to the 0 state shown in FIG. 5. Accordingly, a conventional FeFET based memory cell with only one ferroelectric gate layer can store two states of a single memory bit.

As noted above, the gate 110 of the FeFET 101 illustrated in FIG. 1A has two ferroelectric layers 120a and 120b. FIGS. 8 and 9 illustrate examples of portions of the gate 110, including the first and second ferroelectric layers 120a and 120b, the gate terminals 130 and 134, and the buffer layer 132. In the example of FIG. 8, the buffer layer 132 is formed from metal or another suitable conductive material. FIG. 9 illustrates an alternative embodiment in which the buffer layer 132' is formed of a non-conductive material. As shown in FIG. 8, the voltage applied to the gate terminals 130 and 134 and resulting E field is evenly distributed with the conductive buffer layer 132. With the non-conductive buffer layer 132' shown in FIG. 9, the voltage and resulting E field may curve at the edges of the ferroelectric layer 120a or 120b. In still further embodiments, the buffer layer 132 may be omitted.

Each of the ferroelectric layers 120a and 120b is configured to have a respective unique switching E-field. In the example shown in FIGS. 8 and 9, the surface area of the planar ferroelectric layers 120a and 120b is different, which results in different switching E fields for the ferroelectric layers 120a and 120b. FIG. 10 is a schematic top view of the

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ferroelectric layers 120a and 120b, conceptually illustrating the difference in size or surface area, showing the ferroelectric layer 120a having a smaller area than the ferroelectric layer 120b. The illustrated ferroelectric layers 120a and 120b are planar structures and as such, surface area is varied. However, ferroelectric layers having other shapes and areas are within the scope of the disclosure as long as the ferroelectric layers demonstrate different switching E fields.

In general, the ferroelectric layer 120a or 120b with a smaller area will have a larger voltage drop, and thus have a higher switching E field than the layer with larger area. If the two layers have the same thickness and are formed from the same ferroelectric material, then the one with smaller area will switch first, or at a lower voltage level. To switch the layer with larger area, a higher voltage level applied to the gate terminals is required. As described further below, the two ferroelectric layers 120a and 120b with respective different surface areas A1 and A2 result in a FeFET with multiple threshold voltages.

More particularly, the conductive gate terminals 130, 134 and buffer layer 132 are separated by the dielectric ferroelectric layers 120a and 120b, and thus form capacitors C1 and C2. The different surface areas of the respective ferroelectric layers 120a and 120b results in respective different switching E fields for these ferroelectric layers. As such, the individual ferroelectric layers 120a and 120b of the gate 110 may be individually controlled to provide four different V_t levels as shown in FIG. 11, allowing four different data states to be stored by the multi-bit memory cell 100 including the FeFET 101. FIGS. 12-15 illustrate the polarization of each of the ferroelectric layers 120a and 120b corresponding to the four V_t levels shown in FIG. 11. FIG. 12 illustrates polarization for the V_{t00} threshold voltage, where both of the ferroelectric layers 120a and 120b are in the 0 state. FIG. 13 illustrates polarization for the V_{t01} threshold voltage, where the first ferroelectric layer 120a is in the 0 state and the second ferroelectric layer 120b is in the 1 state. FIG. 14 illustrates polarization for the V_{t10} threshold voltage, where the first ferroelectric layer 120a is in the 1 state and the second ferroelectric layer 120b is in the 0 state. FIG. 15 illustrates polarization for the V_{t11} threshold voltage, where both of the ferroelectric layers 120a and 120b are in the 1 state.

FIG. 16 illustrates an example of a memory device 200 that has an array 202 of the multiple bit memory cells 100 arranged in rows and columns. Each of the rows has a corresponding word line WL. Each of the columns has a corresponding source bit line SBL, a corresponding read bit line RBL, and a corresponding write bit line WBL. A memory controller 210, which may be implemented by any suitable processing device, is configured to control application of signals to the appropriate word line WL and bit lines SBL, RBL and WBL based on a received memory address. An I/O circuit 220 is connected to the bit lines SBL, RBL and WBL to read and write data from the memory cells 100 in response to the controller 210. Operation of the memory device 200 in accordance with disclosed examples is discussed further below.

FIG. 17 illustrates further aspects of the memory device 200, where the multi-bit memory cell 100 includes the FeFET 101 discussed above. FIG. 17 illustrates an example of one of the memory cells 100, along with its connections to the word lines WL and bit lines SBL, RBL and WBL. More particularly, one source/drain terminal 114 of the multi-bit memory cell 100 is connected to the source bit line SBL, while the other source/drain terminal 116 is connected to the read bit line RBL. An access transistor 230 is

connected between the write bit line WBL and the gate terminal **134**. More particularly, each of the memory cells **100** has a respective access transistor **230**, where one source/drain terminal of the access transistor **230** is connected to the gate terminal **134** of its respective multi-bit memory cell **100**, and the other source/drain terminal is connected to the write bit line WBL of the corresponding column. The gate terminal of the access transistor **230** is connected to the word line WL of the corresponding row of the array **202**.

As noted above, the memory cell **100** is a multiple bit memory cell (i.e. able to store multiple bits of data rather than only a single data bit as with conventional memory cells). The multi-bit memory cell **100** includes the FeFET **101**, the gate **110** of which has two ferroelectric layers in the example of FIG. 17, where the ferroelectric layers **120a** and **120b** are configured such that the FeFET **101** has 4 Vt levels, as will be explained further below.

To write to the multi-bit memory cell **100**, the corresponding word line WL is on (i.e. logic 1 or high). In the example shown in FIG. 17, the access transistor **230** is an NMOS transistor, so the high word line signal turns the access transistor **230** on, connecting the write bit line WBL to the gate terminal **134** of the FeFET **101**. For the write operation, the source bit line SBL and the read bit line RBL are connected to the same potential, while the write bit line WBL is connected to another potential with different polarity. The potential difference between the write bit line WBL and the read and source bit lines RBL/SBL is referred to as the write voltage Vw.

Scanning the write voltage Vw from low to high and high to low can produce the four different threshold states denoted as 00, 01, 10, and 11 shown in the transition state diagram of FIG. 11 discussed above. The four different threshold voltages V00, V01, V10, and V11 correspond to the four different data states 00, 01, 10 and 11 of the two-bit memory cell **100**. More particularly, for the FeFET **101** of FIG. 17 that has two ferroelectric layers **120a** and **120b**, the write voltage Vw has one minimum level Vc1 that will cause the first ferroelectric layer **120a** to switch state, and a second minimum level Vc2 that will cause the second ferroelectric layer **120b** to switch state.

As noted above, the total potential difference between the write bit line WBL and the read and source bit lines RBL/SBL is the write voltage Vw. The respective voltage drops across the first and second ferroelectric layers **120a** and **120b** together make up the total write voltage Vw. The respective portions VFe1 and VFe2 of the total write voltage Vw for the ferroelectric layers **120a** and **120b** may be determined according to the respective capacitances of the first and second ferroelectric layers **120a** and **120b** as shown below.

$$V_{Fe1} = Vw * C2 / (C1 + C2) \quad (1)$$

$$V_{Fe2} = Vw * C1 / (C1 + C2) \quad (2)$$

C1 and C2 are capacitances for the respective first and second ferroelectric layers **120a** and **120b** and may be calculated as follows.

$$C1 = \epsilon A1 / d \quad (3)$$

$$C2 = \epsilon A2 / d \quad (4)$$

ϵ is the dielectric constant, A1 and A2 are the areas of the first and second ferroelectric layers **120a** and **120b**, and d is the thickness (i.e. distance between conductive plates) of the first and second ferroelectric layers **120a** and **120b**. In the

illustrated example, the first and second ferroelectric layers **120a** and **120b** are connected in series, and thus the respective capacitances C1 and C2 are in series. However, other connections of the ferroelectric layers **120a** and **120b**, such as parallel connections, are within the scope of the disclosure. Since the dielectric constant E and the thickness d is the same for both of the ferroelectric layers **120a** and **120b**, the capacitance C1 and C2 will vary with the respective areas A1 and A2.

The minimum write voltage Vw levels Vc1 and Vc2 that will cause the respective first and second ferroelectric layers **120a** and **120b** to switch state may be calculated as follows.

$$Vc1 = Ec1 * d * (C1 + C2) / C2 \quad (5)$$

$$Vc2 = Ec2 * d * (C1 + C2) / C1 \quad (6)$$

Ec1 and Ec2 are the switching E fields to write the ferroelectric layers **120a** and **120b** from a 0 to a 1 state. To switch the ferroelectric layers **120a** and **120b** from the 1 to the 0 state, -Vc1 and -Vc2 are applied.

As such, the gate **110** having the two ferroelectric layers **120a** and **120b** with different switching E fields has four different states resulting from different voltage levels applied across the gate electrodes **130** and **134**. Thus, for the double gate FeFET **101** having the two ferroelectric layers **120a** and **120b**, there are four different Vw voltages that cause the ferroelectric layers **120a** and **120b** to switch states: Vw1, Vw2, Vw3 and Vw4. FIG. 18 illustrates relationships among the write voltages Vw1-Vw4 and the switching voltages $\pm Vc1$ and $\pm Vc2$ in accordance with some examples. In the example of FIG. 18, the write voltage Vw1 is greater than Vc1 but less than Vc2 (i.e. $Vc1 < Vw1 < Vc2$), and the write voltage Vw2 is greater than Vc2 (i.e. $Vw1 > Vc2$). Further, the write voltage Vw3 is less than -Vc1 but greater than -Vc2 (i.e. $-Vc2 < Vw3 < -Vc1$), and the write voltage Vw4 is less than -Vc2 (i.e. $Vw4 < -Vc2$).

In some examples, Vc1 is about $\pm 0.25V$, while Vc2 is about $\pm 0.5V$. In some implementations, it may be desirable to calculate the write voltages based on relative dimensions of the ferroelectric layers **120a** and **120b**. For instance, specific write voltages may be calculated based on relative areas A1 and A2 of the first and second ferroelectric layers **120a** and **120b**, respectively. The area A2 of the second ferroelectric layer **120b** may be a factor n (n>0) larger than the area A1 of the first ferroelectric layer **120a**, indicated as follows

$$A2 = n * A1 \quad (7)$$

If the dielectric constant E and thickness (i.e. distance d) of the ferroelectric layers **120a** and **120b** is the same, the capacitance of the ferroelectric layers follows the area relationship.

$$C2 = n * C1 \quad (8)$$

The minimum write voltages Vc1 and Vc2 may then be calculated as shown below.

$$Vc1 = Ec1 * d * (n+1) / n \quad (9)$$

$$Vc2 = Ec2 * d * (n+1) \quad (10)$$

As noted above, E1 and E2 are the E fields for changing the state of (i.e. writing) the respective first and second ferroelectric layers **120a** and **120b**. Thus, if n=2 (i.e. the area of the second ferroelectric layer **120b** is twice that of the first ferroelectric layer **12a**),

$$Vc1 = 1.5 Ec1 * d \quad (11)$$

$$Vc2 = 3 Ec2 * d \quad (12)$$

FIG. 20 is a flow diagram illustrating a general method 150 for writing data to a multi-gate FeFET memory cell, such as the memory cell 100 disclosed above. At step 152, a memory cell is provided, such as the multi-bit memory cell 100. The provided memory cell includes the multi-gate FeFET 101, which includes a source, a drain and a gate. At step 154, a first predetermined signal, such as the Vw1 write voltage, is applied to the gate to write a first data value to a bit (e.g. 01) of the multi-bit memory cell 100. At step 156, a second predetermined signal, such as the Vw2 write voltage, is applied to the gate to write a second bit (e.g. 11) of the memory cell.

More particularly, the possible switching transitions of the ferroelectric layers 120a and 120b for the write voltages Vw1-Vw4 are shown in FIG. 19. In the 00 state, both the first and second ferroelectric layers 120a and 120b are in the 0 state. Applying the write voltage Vw1 (i.e. greater than Vc1 but less than Vc2) to the write bit line WBL results in the first ferroelectric layer 120a changing from the 0 state to the 1 state, while the state of the second ferroelectric layer remains unchanged. Thus, the double gate memory cell 100 in the 00 state transitions to the 01 state. Applying the write voltage Vw2 (i.e. greater than Vc2) to the write bit line WBL results in the second ferroelectric layer 120b changing from the 0 state to the 1 state, so the double gate memory cell 100 in the 01 state transitions to the 11 state. However, since the Vw2 write voltage is higher than the both the Vc1 and Vc2 voltage levels, applying the Vw2 write voltage to the WBL will cause both ferroelectric layers 120a and 120b to transition from 0 to 1. Thus, when the memory cell 100 is in the 00 state, applying the Vw2 voltage to the WBL will cause both ferroelectric layers 120a and 120b to switch to the 11 state, writing the memory cell directly from the 00 state to the 11 state.

Applying the write voltage Vw3 (i.e. less than -Vc1 but greater than -Vc2) to the write bit line WBL results in the first ferroelectric layer 120a changing from the 1 state to the 0 state, while the state of the second ferroelectric layer remains unchanged. Thus, the double gate memory cell 100 in the 11 state transitions to the 10 state. Applying the write voltage Vw4 (i.e. less than -Vc2) to the write bit line WBL results in the second ferroelectric layer 120b changing from the 1 state to the 0 state, so the double gate memory cell 100 in the 10 state transitions to the 00 state. However, since the Vw4 write voltage is lower than the both the -Vc1 and -Vc2 voltage levels, applying the Vw4 write voltage to the WBL will cause both ferroelectric layers 120a and 120b to transition from 1 to 0. Thus, when the memory cell 100 is in the 11 state, applying the Vw4 voltage to the WBL will cause both ferroelectric layers 120a and 120b to switch to the 00 state, writing the memory cell directly from the 11 state to the 00 state.

Moreover, applying the Vw3 write voltage to the WBL when the memory cell 100 is in the 01 state will cause the first ferroelectric layer 120a to transition to the 0 state, thus writing the memory cell 100 from the 01 state to the 00 state. Similarly, applying the Vw1 write voltage to the WBL when the memory cell 100 is in the 10 state will cause the first ferroelectric layer 120a to transition to the 1 state, thus writing the memory cell 100 from the 10 state to the 11 state.

Referring back to FIG. 20, the method 150 further includes applying a third predetermined signal to the gate at step 158 for a data read process. In step 160, a signal between the FeFET source and drain is determined, and the data stored in the memory cell 100 is determined based thereon in step 162.

More particularly, to read data from the double gate memory cell 100, the signal on the wordline WL is high, turning on the access transistor 230. A read voltage Vread is applied to the write bit line WBL and thus to the gate 110 of the FeFET 101. In some examples, the read voltage Vread is between -Vc1 and Vc1 as shown in FIG. 18. A voltage or current passing between the read bit line RBL and the source bit line SBL will vary according to the four different Vt states (see FIG. 11). Reading the resulting current or voltage using a multiple bit sense amplifier included in the I/O circuit 220 derives the 2 bits of data from the double gate memory cell 100.

As discussed above, the unique switching E fields for the ferroelectric layers 120a and 120b may be achieved by varying the surface area of the ferroelectric layers 120a and 120b. Alternative embodiments may achieve unique switching E fields for each of the ferroelectric layers 120a and 120b by varying other factors. For example, in some embodiments the area of the ferroelectric layers 120a and 120b may be the same while the dielectric constant of the ferroelectric layers 120a and 120b is varied to achieve respective different switching E fields for the ferroelectric layers 120a and 120b.

In general, a ferroelectric layer having a lower dielectric constant will have a larger voltage drop, and thus have a higher switching E field than a ferroelectric layer having a larger dielectric constant. For example, if the two ferroelectric layers 120a and 120b have the same thickness d (i.e. vertical dimension as shown in FIGS. 8 and 9) and also have the same surface area, but have different dielectric constants, the layer having the lower dielectric constant will switch first—i.e., at a lower write voltage Vw. To switch the layer with the higher dielectric constant, a stronger voltage is applied across the gate terminals 130 and 134. Since the different ferroelectric layers switch in response to different applied voltages (i.e. resulting from different switching E fields), the multi-gate FeFET 101 will demonstrate multiple threshold voltages as described above.

FIG. 21 illustrates a generalized case of a memory cell 300 in accordance with disclosed examples. As with the embodiment shown in FIG. 1, the memory cell 300 includes a multi-gate FeFET 301 having N ferroelectric layers (N is a positive integer greater than 1), including ferroelectric layers 120a, 120b . . . 120N. The example shown in FIG. 21 has the N ferroelectric layers arranged in a stacked configuration with the size (i.e. surface area) of the ferroelectric layers getting larger from top to bottom. Other configurations are within the scope of this disclosure.

In the FeFET 101 of the multi-bit memory cell 100 shown in FIG. 1, N=2. As with the FeFET 101 shown in FIG. 1, the gate 110 of the FeFET 301 shown in FIG. 21 includes a plurality of ferroelectric layers configured such that each of the ferroelectric layers has a respective unique switching E-field. The ferroelectric layers are arranged in a stacked arrangement, and a gate metal layer 130 abuts one side of the bottommost ferroelectric layer 120N and is connected to a gate electrode 126 by a via 128. Buffer layers 132 are situated between the each of the adjacent ferroelectric layers 120a, 120b N, and a conductive gate terminal 134 is situated on the uppermost ferroelectric layer 120a.

As with the multi-bit memory cell 100, the memory cell 300 has a substrate 112 that may be, for example, a semiconductor layer of a semiconductor-on-insulator (SOI) substrate or a bulk semiconductor substrate. The FeFET 301 comprises a pair of source/drain regions 114, 116 that have a first doping type and are arranged within the substrate 112, respectively on opposite sides of a channel region 118. The

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channel region **118** has a second doping type opposite the first doping and is arranged in the substrate **112** laterally between the source/drain regions **114**, **116**. The first and second doping types may, for example, respectively be n-type and p-type, or vice versa.

The gate **110** has a gate metal layer **126** arranged over the substrate **112** laterally between the source/drain regions **114**, **116**, and is spaced from the substrate **112** by a non-ferroelectric gate oxide **124**. In some examples, the non-ferroelectric gate oxide material is silicon dioxide. The ferroelectric layers **120a**, **120b** . . . **120N** may alternatively be formed directly over the non-ferroelectric gate oxide **124** to form a composite gate structure. The gate electrode **126** is conductive and may comprise metal, doped polysilicon, or a combination thereof. In the illustrated embodiment, the gate electrode **126** is electrically coupled to the ferroelectric layers by a conductive via **128**.

As shown in FIG. **22**, the generalized multi-bit FeFET **301** having N ferroelectric layers will exhibit 2N threshold voltages V_t . Further, the generalized multi-bit FeFET **301** having N ferroelectric layers has a maximum number of data bits according to $\log_2(2N)$. The example discussed above in conjunction with FIG. **1A** has two ferroelectric layers **120a** and **120b**, and thus $N=2$. Accordingly, the FeFET **101** exhibits four (i.e. $2N=4$) threshold voltages V_t as shown in FIG. **11**, and can write two data bits (i.e. $\log_2(2N)=2$).

FIGS. **23-25** illustrate aspects of another example memory cell **310** applying the general case **300** shown in FIG. **21**. The memory cell **310** includes a multi-bit FeFET **311** having 4 ferroelectric layers **120a-120d**. In other words, $N=4$. As with the earlier described embodiments, the four ferroelectric layers **120a-120d** are arranged in a stacked configuration with the size (i.e. surface area) of the ferroelectric layers getting larger from top to bottom. Other configurations are within the scope of this disclosure.

As with the previous examples, the FeFET **311** shown in FIG. **23** has ferroelectric layers **120a-120d** configured such that each has a respective unique switching E-field. The FeFET **311** of the memory cell **310** has one source/drain terminal **114** connected to the source bit line SBL, while the other source/drain terminal **116** is connected to the read bit line RBL. An access transistor **230** is connected between the write bit line WBL and the gate terminal **134**. More particularly, one source/drain terminal of the access transistor **230** is connected to the gate terminal **134** of its respective memory cell **310**, and the other source/drain terminal is connected to the write bit line WBL of the corresponding column. The gate terminal of the access transistor **230** is connected to the word line WL of the corresponding row of the array **202**.

As noted above, the multi-bit FeFET **311** having N ferroelectric layers will exhibit 2N threshold voltages V_t . Accordingly, the memory cell **310** having the FeFET **311** has eight V_t levels, and thus has eight minimum write voltages levels $\pm V_{c1}-V_{c4}$ (i.e., the voltages that cause the ferroelectric layers **120a-120d** to switch between the 0 and 1 state). FIG. **24** illustrates relationships among the switching voltages $\pm V_{c1}$, $\pm V_{c2}$, $\pm V_{c3}$ and $\pm V_{c4}$ and write voltages $V_{w1}-V_{w8}$ for the memory cell **310**. In the example of FIG. **24**, the write voltage V_{w1} is greater than V_{c1} but less than V_{c2} (i.e. $V_{c1} < V_{w1} < V_{c2}$), the write voltage V_{w2} is greater than V_{c2} but less than V_{c3} (i.e. $V_{c2} < V_{w2} < V_{c3}$), the write voltage V_{w3} is greater than V_{c3} but less than V_{c4} (i.e. $V_{c3} < V_{w3} < V_{c4}$), and the write voltage V_{w4} is greater than V_{c4} (i.e. $V_{w4} > V_{c4}$). Further, the write voltage V_{w5} is less than $-V_{c1}$ but greater than $-V_{c2}$ (i.e. $-V_{c2} < V_{w5} < -V_{c1}$), the write voltage V_{w6} is less than $-V_{c2}$ but greater

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than $-V_{c3}$ (i.e. $-V_{c3} < V_{w6} < -V_{c2}$), the write voltage V_{w7} is less than $-V_{c3}$ but greater than $-V_{c4}$ (i.e. $-V_{c4} < V_{w7} < -V_{c3}$), and the write voltage V_{w8} is less than $-V_{c4}$ (i.e. $V_{w8} < -V_{c4}$).

The $V_{c1}-V_{c4}$ voltages may be calculated as follows.

$$V_{c1} = E_{c1} * d * (C1C2C3 + C1C2C4 + C1C3C4 + C2C3C4) / C1C3C4 \quad (13)$$

$$V_{c2} = E_{c2} * d * (C1C2C3 + C1C2C4 + C1C3C4 + C2C3C4) / C1C3C4 \quad (14)$$

$$V_{c3} = E_{c3} * d * (C1C2C3 + C1C2C4 + C1C3C4 + C2C3C4) / C1C2C4 \quad (15)$$

$$V_{c4} = E_{c4} * d * (C1C2C3 + C1C2C4 + C1C3C4 + C2C3C4) / C1C2C3 \quad (16)$$

E_{c1-4} are the respective switching E fields to write the ferroelectric layers **120a-120d** from a 0 to a 1 state, $C1-C4$ are the capacitances of the ferroelectric layers **120a-12d** and d is the ferroelectric layer thickness.

If $C4=4C1$, $C3=3C1$, $C2=2C1$, then the minimum $V_{c1}-V_{c4}$ voltage levels may be calculated according to equations (9) and (10) as follows

$$V_{c1} = 50/24 * E_{c1} * d$$

$$V_{c2} = 2 * V_{c1}$$

$$V_{c3} = 3 * V_{c1}$$

$$V_{c4} = 4 * V_{c1}$$

Switching transitions of the ferroelectric layers **120a-120d** for the write voltages $V_{w1}-V_{w8}$ are shown in FIG. **25**. In the 0000 state, all of the ferroelectric layers **120a-120d** and **120b** are in the 0 state. Applying the write voltage V_{w1} (i.e. greater than V_{c1} but less than V_{c2}) to the write bit line WBL results in the first ferroelectric layer **120a-120d** changing from the 0 state to the 1 state, while the states of the ferroelectric layers **120b-120d** remain unchanged. Thus, the quadruple gate memory cell **310** in the 0000 state transitions to the 0001 state. Applying the write voltage V_{w2} (i.e. greater than V_{c1} and V_{c2} but less than V_{c3}) to the write bit line WBL results in the second ferroelectric layer **120b** changing from the 0 state to the 1 state, so the memory cell **310** in the 0001 state transitions to the 0011 state. However, since the V_{w2} write voltage is higher than the both the V_{c1} and V_{c2} voltage levels, applying the V_{w2} write voltage to the WBL will cause both ferroelectric layers **120a** and **120b** to transition from 0 to 1. Thus, when the memory cell **310** is in the 0000 state, applying the V_{w2} voltage to the WBL will cause both ferroelectric layers **120a** and **120b** to switch to the 11 state, writing the memory cell directly from the 0000 state to the 0011 state.

Applying the write voltage V_{w3} (i.e. greater than V_{c1} , V_{c2} and V_{c3} but less than V_{c4}) to the write bit line WBL results in the third ferroelectric layer **120c** changing from the 0 state to the 1 state, while the state of the ferroelectric layer **120d** remains unchanged. Thus, the memory cell **310** in any of the 0000, 0001, 0011 states will transition to the 0111 state. Applying the write voltage V_{w4} (i.e. greater than V_{c1} , V_{c2} , V_{c3} and V_{c4}) to the write bit line WBL results in the ferroelectric layer **120d** changing from the 0 state to the 1 state, so the memory cell **310** in any of the 0000, 0001, 0011, and 0111 states transitions to the 1111 state.

Applying the write voltage V_{w5} (i.e. less than $-V_{c1}$ but greater than $-V_{c2}$) to the write bit line WBL results in the first ferroelectric layer **120a** changing from the 1 state to the 0 state, while the states of the ferroelectric layers **120b-120d**

remains unchanged. Thus, the memory cell **310** in the 1111 state transitions to the 1110 state. Applying the write voltage **Vw6** (i.e. less than $-Vc1$ and $-Vc2$ but greater than $-Vc3$) to the write bit line **WBL** results in the second ferroelectric layer **120b** changing from the 1 state to the 0 state, so the memory cell **310** in the 1110 state transitions to the 1100 state. Since the **Vw6** write voltage is less than the both the $-Vc1$ and $-Vc2$ voltage levels, applying the **Vw6** write voltage to the **WBL** will cause both ferroelectric layers **120a** and **120b** to transition from 1 to 0, so that when the memory cell **310** is in the 1111 state, applying the **Vw6** voltage to the **WBL** will cause both ferroelectric layers **120a** and **120b** to switch to the 0 state, writing the memory cell directly from the 1111 state to the 1100 state.

Applying the write voltage **Vw7** (i.e. less than $-Vc1$, $-Vc2$ and $-Vc3$ but greater than $-Vc4$) to the write bit line **WBL** results in the third ferroelectric layer **120c** changing from the 1 state to the 0 state, while the state of the ferroelectric layer **120d** remains unchanged. Thus, the memory cell **310** in any of the 1111, 1110, 1100 states will transition to the 1000 state. Applying the write voltage **Vw8** (i.e. less than $-Vc1$, $-Vc2$, $-Vc3$ and $-Vc4$) to the write bit line **WBL** results in the ferroelectric layer **120d** changing from the 1 state to the 0 state, so the memory cell **310** in any of the 1000, 1110, 1100, and 1000 states transitions to the 0000 state.

Reading data from the quadruple gate memory cell **310** follows the same process as described above for the double gate memory cell **100**. The signal on the word line **WL** is high, turning on the access transistor **230**. A read voltage **Vread** is applied to the write bit line **WBL** and thus to the gate **110** of the FeFET **311**. In some examples, the read voltage **Vread** is between $-Vc1$ and $Vc1$ as shown in FIG. **24**. A voltage or current passing between the read bit line **RBL** and the source bit line **SBL** will vary according to the eight different V_t states of the memory cell **310**. Reading the resulting current or voltage using a multiple bit sense amplifier included in the I/O circuit **220** derives the 3 bits of data from the quadruple gate memory cell **310**.

Disclosed embodiments provide a non-volatile memory cell that is configured to store multiple bits of data, thus reducing space requirements for a memory device. Such multi-bit memory cells include a multiple gate FeFET structure that has a plurality of ferroelectric layers that each exhibit a respective unique switching E-field. Accordingly, the multi-bit memory cell exhibits more threshold voltage V_t values than a conventional FeFET memory cell.

More particularly, disclosed embodiments include a memory device having a plurality of memory cells. Each memory cell includes a multi-gate FeFET that has a first source/drain terminal, a second source/drain terminal, and a gate with a plurality of ferroelectric layers configured such that each of the ferroelectric layers has a respective unique switching E-field.

In accordance with further embodiments, a memory device includes an array of multi-bit memory cells arranged in rows and columns. Each of the rows has a corresponding word line, and each of the columns has a corresponding source bit line, a corresponding read bit line, and a corresponding write bit line. A plurality of access transistors each have a first source/drain terminal connected to a respective one of the multi-bit memory cells, a second source/drain terminal connected to the write bit line of the corresponding column, and a gate terminal connected to the word line of a corresponding row. Each of the multi-bit memory cells includes a FeFET having a first source/drain terminal coupled to the source bit line of the corresponding column,

a second source/drain terminal coupled to the read bit line of the corresponding column, and a gate comprising N ferroelectric layers (N is a positive integer greater than 1). The N ferroelectric layers are configured such that the FeFET has $2N$ threshold voltage (V_t) levels.

In accordance with still further examples, a method includes providing a memory cell including a multi-gate FeFET that has a source, a drain and a gate. A first predetermined signal is applied to the gate to write a first data value to a first bit of the memory cell. A second predetermined signal is applied to the gate to write the first data value to a second bit of the memory cell.

This disclosure outlines various embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A memory device including a plurality of memory cells, each memory cell comprising:

a multi-gate ferroelectric gate Field Effect Transistor (FeFET) forming a plurality of ferroelectric layer, the multi-gate FeFET configured such that each of the plurality of ferroelectric layers has a respective unique switching E-field, the multi-gate FeFET further comprising:

a substrate defining first and second source/drain regions having a first doping type, and a channel region having a second doping type opposite the first doping type situated between the first and second source/drain regions;

a gate electrode arranged over the channel region including a non-ferroelectric gate oxide positioned between the gate electrode and the substrate;

a back end of the line (BEOL) stacked ferroelectric gate structure over the gate electrode.

2. The memory device of claim 1, wherein the stacked ferroelectric gate structure comprises a plurality of ferroelectric layers, and each of the ferroelectric layers has one of a different surface area or a different dielectric constant from the others of the ferroelectric layers.

3. The memory device of claim 1, further comprising a buffer layer between a second side of a first conductive gate layer and a first side of a second ferroelectric layer, wherein the buffer layer is electrically conductive.

4. The memory device of claim 1, further comprising a buffer layer between a second side of a first conductive gate layer and a first side of a second ferroelectric layer, wherein the buffer layer is electrically non-conductive insulating.

5. The memory device of claim 2, wherein the ferroelectric layers are electrically connected in series.

6. The memory device of claim 2, wherein the plurality of ferroelectric layers comprises N ferroelectric layers, where N is a positive integer, and wherein the FeFET has $2N$ threshold voltage (V_t) levels.

7. The memory device of claim 1, further comprising: a first source/drain terminal electrically connected to the first source/drain region; and

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a second source/drain terminal electrically connected to the second source/drain region.

8. A memory cell comprising:

a multi-gate ferroelectric gate Field Effect Transistor (FeFET) forming a plurality of ferroelectric layers, the multi-gate FeFET configured such that each of the plurality of ferroelectric layers has a respective unique switching E-field, the multi-gate FeFET further comprising:

a substrate defining first and second source/drain regions having a first doping type, and a channel region having a second doping type opposite the first doping type situated between the first and second source/drain regions;

a gate electrode arranged over the channel region including a non-ferroelectric gate oxide positioned between the gate electrode and the substrate;

a back end of the line (BEOL) stacked ferroelectric gate structure over the gate electrode.

9. The memory cell of claim 8, wherein the stacked ferroelectric gate structure comprises a plurality of ferroelectric layers, and each of the ferroelectric layers has a different surface area from the others of the ferroelectric layers.

10. The memory cell of claim 8, wherein the stacked ferroelectric gate structure comprises a plurality of ferroelectric layers, and each of the ferroelectric layers has a different dielectric constant from the others of the ferroelectric layers.

11. The memory cell of claim 8, further comprising a buffer layer between a second side of a first conductive gate layer and a first side of a second ferroelectric layer, wherein the buffer layer is electrically conductive.

12. The memory cell of claim 8, further comprising a buffer layer between a second side of a first conductive gate layer and a first side of a second ferroelectric layer, wherein the buffer layer is electrically non-conductive insulating.

13. The memory cell of claim 8, wherein the ferroelectric layers are electrically connected in series.

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14. The memory cell of claim 8, wherein the plurality of ferroelectric layers comprises N ferroelectric layers, where N is a positive integer, and wherein the FeFET has 2N threshold voltage (Vt) levels.

15. The memory cell of claim 8, further comprising:

a first source/drain terminal electrically connected to the first source/drain region; and

a second source/drain terminal electrically connected to the second source/drain region.

16. A multi-gate ferroelectric gate Field Effect Transistor (FeFET) arranged in a substrate and capable of storing four distinct threshold voltage (Vt) states, the FeFET comprising:

a source-drain conduction region;

at least two capacitors separated from the source-drain region by the via, each of the at least two capacitors comprising:

a buffer layer,

a conductive gate terminal,

a dielectric ferroelectric layer arranged between the buffer layer and the conductive gate terminal, and

a non-ferroelectric gate oxide positioned between the conductive gate terminal and the substrate,

wherein each of the at least two capacitors has a respective unique switching E-field such that applying a predetermined signal to through the via stores a Vt of the four distinct Vt states in one of a plurality of layers of the multi-gate FeFET corresponding to the unique switching E-field.

17. The memory device of claim 16, wherein each of the ferroelectric layers has one of a different surface area or a different dielectric constant.

18. The memory device of claim 16, wherein the ferroelectric layers are electrically connected in series.

19. The memory device of claim 16, wherein the plurality of ferroelectric layers comprises N ferroelectric layers, where N is a positive integer, and wherein the FeFET has 2N threshold voltage (Vt) levels.

20. The memory device of claim 16, further comprising: a via coupled to the source-drain conduction region.

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